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Intra-day seasonality and abnormal returns in the Brent crude oil futures market

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We investigate intra-day seasonal patterns in Brent crude oil futures traded at the Intercontinental Exchange in London. Our data cover tick data for futures of various maturities from January 2010 to October 2021, a database covering 130 Gigabytes of oil transactions. We convert these data into one-minute data and observe statistically significant intra-day seasonal patterns with peaks and bottoms at particular times of the day, depending on maturities and whether or not spreads are considered. In the second part of our analysis, we explore whether these systematic patterns can be exploited to create arbitrage-like long-short strategies, going long and short at particular times during the day. The answer is yes; even when accounting for realistic transaction costs and margin requirements, some of the proposed strategies can create consistent positive and significant CAPM alphas.

Keywords: Brent crude oil; Futures; Seasonality; High-frequency trading; Intra-day patterns; Abnormal returns; Arbitrage; Large data

1. Introduction

Intra-day seasonality in finance refers to the systematic patterns observed in trading activity, price movements, and volatility that occur within a single trading day. This phenomenon is characterized by distinct fluctuations in market behavior at different times of the day, often influenced by factors such as trading volume, market liquidity, and investor behavior. Understanding intra-day seasonality is crucial for traders and market participants as it can significantly impact trading strategies and risk management.

One of the key aspects of intra-day seasonality is the U-shaped pattern often observed in trading volume and volatility. Research has shown that trading activity tends to be higher at the market open and close, with a noticeable dip in the middle of the trading day. For instance, studies have documented that informed traders are more active during these peak periods, leading to increased liquidity and volatility at the start and end of the trading session (Blau *et al.* 2009, Ryu 2011, Tannous *et al.* 2013). This pattern is not only prevalent in equity markets but also extends to other asset

classes, such as commodities and currencies, where similar intra-day patterns have been identified (Xu *et al.* 2020, Ahmed *et al.* 2021). For markets open 24 h, such as the Intercontinental Exchange (ICE) in London, there is a different intra-day pattern, as our study shows. In such markets, the trading patterns tend to follow the major global financial markets' regular office hours.

The impact of intra-day seasonality on market risk cannot be understated. Models that account for intra-day volatility patterns, such as GARCH and its variations, have been developed to better capture the dynamics of intra-day returns. These models highlight the importance of removing intra-day seasonality to accurately assess market risk and volatility (Giot 2000, 2005, Diao and Tong 2015). Intraday seasonality can lead to mis-estimations of risk if not correctly accounted for, particularly in high-frequency trading environments where small time frames are analyzed (Tian and Guo 2007).

The concept of liquidity seasonality is supported by empirical evidence showing that liquidity levels, often measured by bid-ask spreads, exhibit significant differences at various times during the trading day. Xu and Li (2018) emphasize that liquidity in the Chinese agricultural futures market demonstrates a U-shaped pattern, indicating that liquidity is

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generally higher during the market opening and closing compared to midday. This pattern is consistent across various asset classes, including equities and commodities, where similar intraday liquidity dynamics have been observed (Aradhyula and Ergün 2004, Tissaoui 2012).

The role of information asymmetry and trading strategies complicates the understanding of intra-day seasonality. Research indicates that the timing of trades by informed investors can create significant intra-day patterns, as these traders often execute their trades to minimize the impact of their information on market prices (Saad and Darat 2007, Tannous *et al.* 2013). This behavior contributes to the observed fluctuations in bid-ask spreads and market depth throughout the trading day, reinforcing the need for traders to be aware of these patterns when formulating their strategies (Visaltanachoti *et al.* 2003, Hsieh *et al.* 2020).

Based on the above, intra-day seasonality is a multifaceted phenomenon that encompasses various aspects of market behavior, including trading volume, volatility, and the influence of informed trading. By recognizing and analyzing these patterns, market participants can enhance their trading strategies and improve their risk management practices. But is it possible to earn abnormal returns from the observed intra-day seasonal patterns? Our study investigates this question using high-frequency data for Brent Crude oil futures obtained from the ICE in London. The Brent Crude futures contract is a global benchmark in oil trading and is used to price over three-quarters of the world's traded oil. In our previous study Ewald *et al.* (2022b), we also looked at Brent crude oil futures and found annual seasonal patterns and price anomalies. There, we argued that hedging pressure within the short (oil producer) and long (oil consumer) positions shifts according to an annual cycle, which can explain the findings. In equity futures, such as S&P500 index futures, there is no clear alignment of short and long positions to different groups of investors, and hedging pressure plays a much less prominent role. In Ewald *et al.* (2022b) and Størdal *et al.* (2023), we also looked at natural gas and electricity, but found that the effect in Brent crude oil is the strongest. In conclusion, we focus on Brent crude oil, where our current study extends this agenda within an intraday framework.

The idea that seasonality in financial markets can lead to short-term arbitrage opportunities and abnormal returns is well-established. For instance, research has highlighted the 'January effect', where stock prices typically increase in January, leading to potential trading strategies that exploit this seasonal pattern (Enow 2024, Siklos 2008). Furthermore, the presence of derivative markets, such as futures contracts, allows investors to hedge against seasonal price movements and exploit arbitrage opportunities more effectively (Faff and McKenzie 2002, Bohl *et al.* 2010). The inefficiencies in these markets, often exacerbated by transaction costs and investor behavior, can lead to persistent seasonal anomalies that savvy traders can exploit (Oprea and Țilică 2014). Of particular relevance in our context is recent work by Ewald *et al.* (2022b) who designed a trading strategy that exploits annual seasonal patterns in the risk premia through appropriate long-short positions in crude oil and natural gas futures contracts and delivers abnormal returns in terms of positive CAPM alpha,

even after accounting for realistic transaction costs and margin requirements. In the current paper, we show that their ideas can be used to exploit intra-day seasonal patterns in Brent crude oil markets and create a daily long-short strategy that produces a positive alpha.

2. Data analysis

Our analysis is based on high-frequency data of Brent Crude oil futures from January 2010 to October 2021. For liquidity and representativeness, we focus on contracts with maturities of one month and three months in our analysis. The raw data consists of tick-by-tick, transaction-level observations of all trades carried out on the Exchange (not just oil). Extensive data wrangling was required before the data could be analyzed effectively. The data preparation involved data cleaning, transformation, and restructuring to ensure the dataset contained only the relevant Brent Crude futures contracts and was suitable for subsequent analysis. The raw dataset amounted to 22 terabytes (TB) of Commodities Level 1 data, corresponding to an average of 3 gigabytes (GB) of data with approximately 47 million daily rows. The data was structured in a multi-record CSV format, including header-, quote-, transaction-, and status rows. Figure 1 shows tick-data within a one-minute interval on a given day during the period investigated.

Throughout this process, it was necessary to account for several changes in data structure over time. This meticulous preparation ensured the dataset was clean, organized, and consistent with the required analytical standards for further research. From the raw Brent Crude oil data (approximately 130 GB), we constructed equally-spaced one-minute data using the previous-tick interpolation method. The method involves taking the last available observation before the minute of interest. Other approaches to construct equally spaced data from raw, transaction-level data have been proposed in the literature, but (Hansen and Lunde 2006) argue convincingly for choosing the previous-tick approach, and many researchers, including ourselves, have followed this guidance.

Table 1 presents the descriptive statistics for the 1-minute futures prices of F1, F3, and the spread. F1 represents contracts with a maturity of one month, while F3 represents contracts with a maturity of three months. While longer maturities than three months are traded, they are not traded in sufficient liquidity at the 1-minute level and can, therefore, not be facilitated in our analysis. Figure 2 shows the liquidity as measured by the average number of trades per hour per day of contracts with maturities of 1, 3, 6, and 12 months.

We define the spread as the price difference between F1 and F3, i.e. $spread = F1 - F3$. Figure 3 illustrates the price dynamics of the futures and the spread. The overall trend of F1 and F3 is similar, and the spread typically fluctuates within the range of $[-5, 5]$ dollars.

Trading at the ICE starts at 1:00 in the morning and ends at 22:59 at night UK time each trading day. To better identify the intra-day seasonal pattern, we calculate the average price per minute and per hour across multiple trading

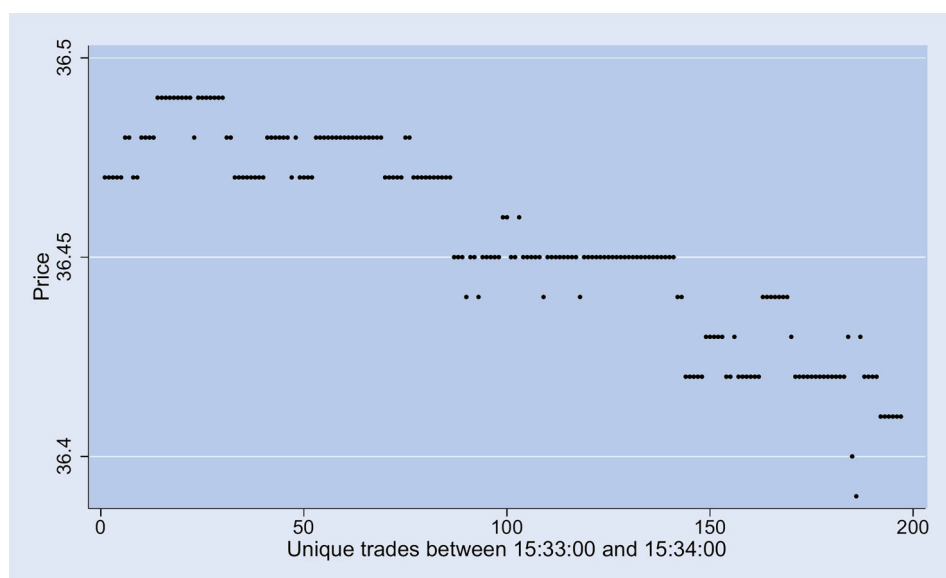


Figure 1. Tick-data within a one-minute interval on a given day during the period January 2010 to October 2021.

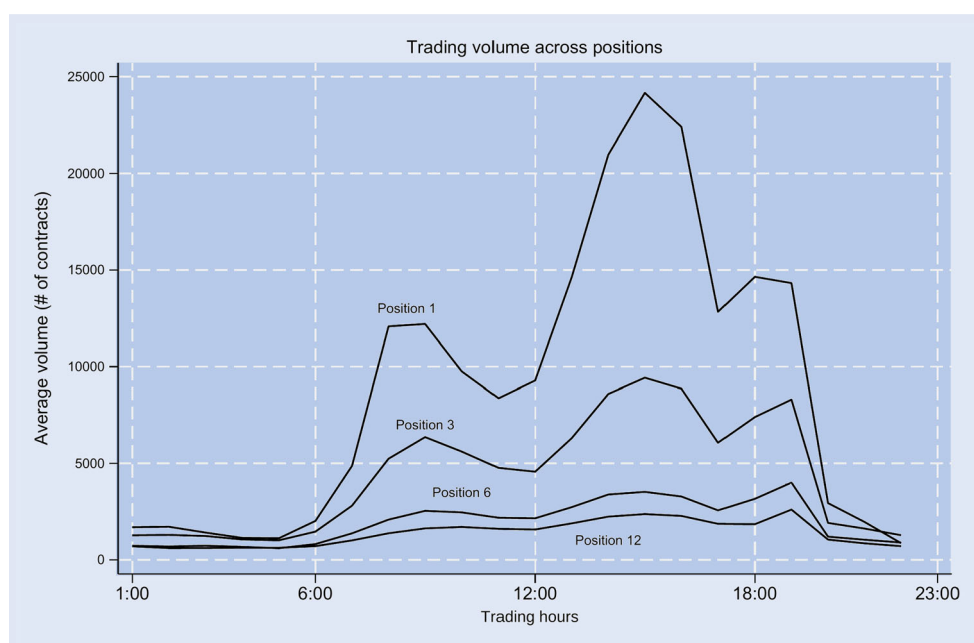


Figure 2. Liquidity as measured by average number of trades per hour per day of contracts with maturities 1, 3, 6, and 12 months.

Table 1. Descriptive statistics, 04/01/2010-29/10/2021.

	Mean	Median	Maximum	Minimum	Std. Dev.	Skewness	Kurtosis	Observations
F1	76.06	71.01	128.06	15.99	26.04	0.22	-1.21	3 985 080
F3	76.00	70.71	126.21	20.00	25.53	0.22	-1.23	3 985 080
Spread	0.06	0.09	8.95	-6.88	0.91	0.01	2.15	3 985 080

Note: Descriptive statistics are provided for the 1-minute futures prices of F1, F3, and the spread. F1 represents contracts with a maturity of one month, while F3 represents contracts with a maturity of three months. The spread is defined as the price difference between F1 and F3, i.e. $spread = F1 - F3$.

days, as shown in figures 4 and 5, respectively. A clear intra-day seasonal pattern can be observed in the futures prices and the spread. The F1 price peaks at 11:00 a.m. and hits a low at 5:00 p.m.; similarly, the F3 price peaks at 11:00 a.m. but reaches its lowest point at 4:00 p.m. The spread reaches its highest value at 10:00 p.m. and its lowest at 6:00 p.m.

To further examine the presence of intraday seasonal patterns, boxplots and histograms are used to visually inspect the distribution of these price differences[†], see figure 6. The

[†] F1: Hour 11 (11:00 AM) minus Hour 17 (5:00 PM); F3: Hour 11 (11:00 AM) minus Hour 16 (4:00 PM); Spread: Hour 22 (10:00 PM) minus Hour 18 (6:00 PM).

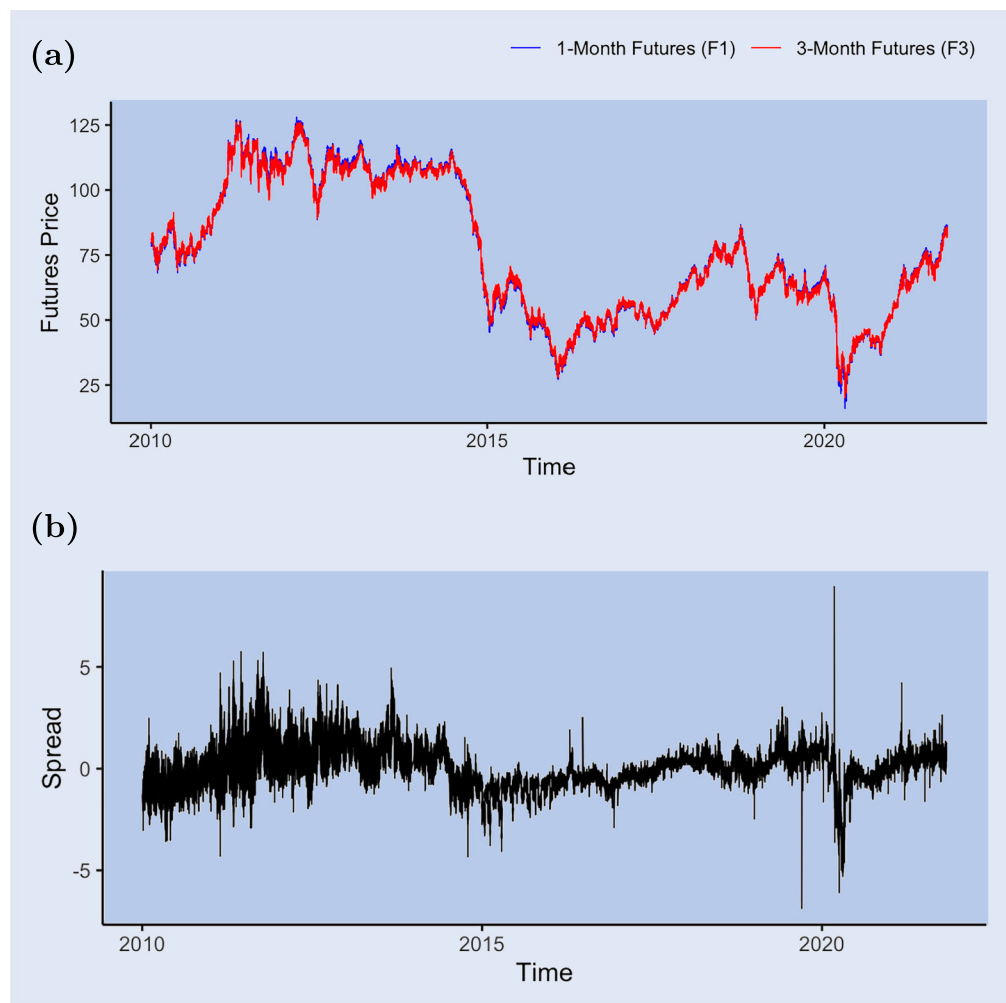


Figure 3. Price over time: (a) F1 and F3; (b) Spread.

box plots illustrate central tendencies, variances, and potential outliers, while the histograms provide insights into the frequency distribution of observed price differences. The box plots reveal that the median price differences in all three cases are positive. The absence of overlap with zero in the interquartile ranges indicates that these differences are systematically positive. Similarly, the histograms confirm that the majority of observed price differences cluster around positive values, further suggesting a consistent intraday seasonal pattern.

To rigorously test the significance of these observed intraday patterns, the Wilcoxon signed-rank test and paired t-test are employed. The Wilcoxon signed-rank test is a non-parametric test that assesses whether the median of paired differences significantly deviates from zero, accounting for potential non-normality in data distribution. Paired observations ensure a valid comparison of intraday prices within the same trading day. Results of the Wilcoxon signed-rank test are presented in table 2. The test statistics confirm that the median intraday price differences are significantly greater than zero at the conventional 5% significance level for all pairs tested.

Complementing the Wilcoxon test, the paired t-test is conducted to verify if the mean differences between intraday peaks and lows are significantly positive. This parametric test assumes the normality of the data and evaluates the significance of mean-level differences. Results are summarized

Table 2. Wilcoxon signed-rank test results.

	V-Statistic	<i>p</i> -value
F1: Hour 11 > Hour 17	2 367 408	0.03299
F3: Hour 11 > Hour 16	2 385 792	0.01313
Spread: Hour 22 > Hour 18	2 475 902	3.835e – 06

Table 3. Paired t-Test results.

	t-statistic	<i>p</i> -value	Mean Difference
F1: Hour 11 > Hour 17	2.2958	0.01088	0.03583
F3: Hour 11 > Hour 16	2.6874	0.00362	0.03804
Spread: Hour 22 > Hour 18	3.8965	4.985e – 05	0.02719

in table 3. The paired t-test results corroborate those of the Wilcoxon test, demonstrating significant positive mean differences, confirming that intraday peak prices consistently exceed the corresponding lows.

In summary, the combination of descriptive analysis and statistical testing provides robust evidence for the existence of significant intraday seasonal price differences. These findings establish a solid empirical foundation for the trading strategy that capitalizes on such predictable intraday patterns.

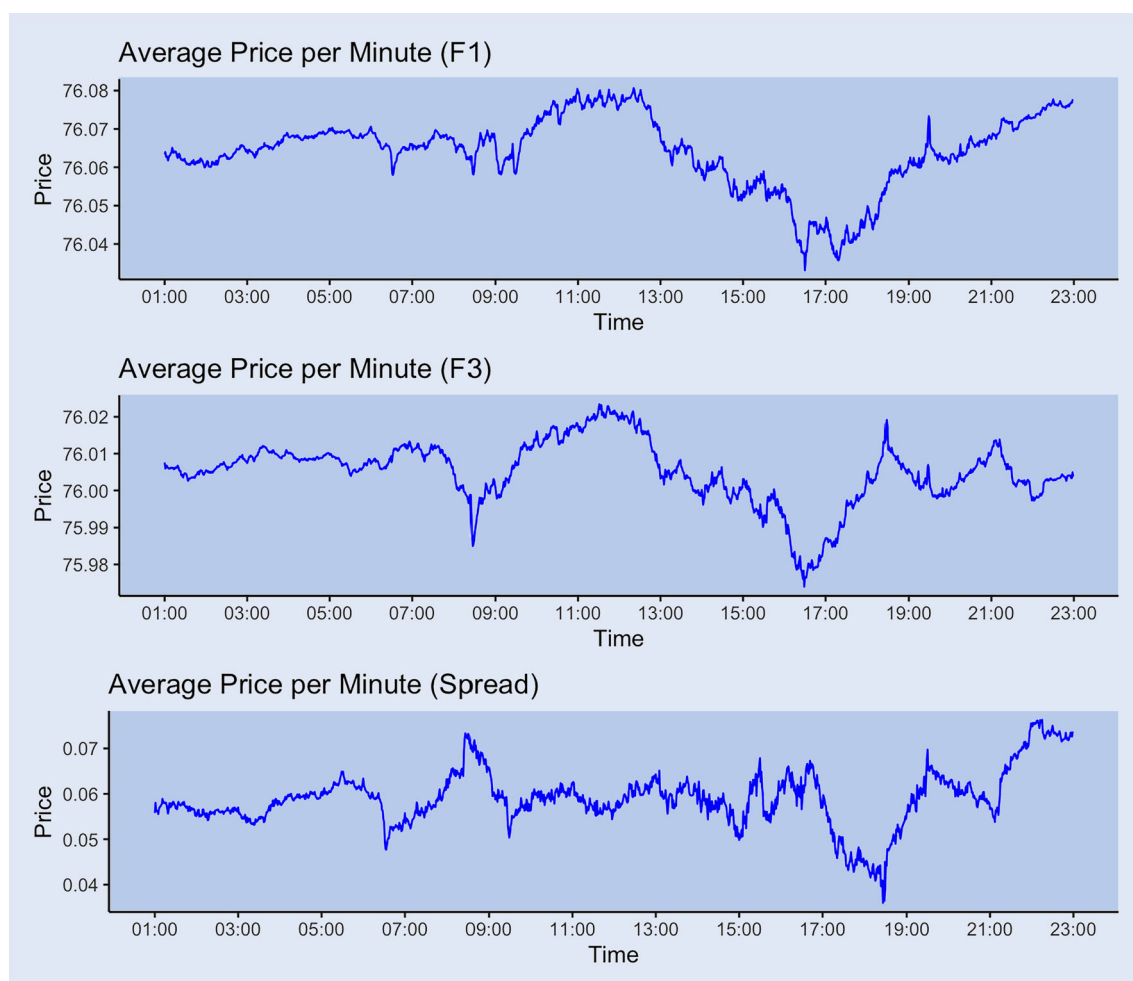


Figure 4. Intraday average price per minute across trading days.

3. Trading strategy

Having detected a seasonal pattern in the F1 and F3 futures prices, as well as the spread, we develop a simple and naïve trading strategy based on these observations. The strategy involves going long on the contract when the price is low and shorting the contract when the price is high. We consider two trading approaches: the first uses a single contract, while the second incorporates both contracts, accounting for the spread. While the margins between the peak and bottom intra-day futures prices are relatively low, i.e. within a range of a couple of cents, given that the difference is highly significant, these can be scaled up sufficiently by entering a large enough position with very low risk.

For the purpose of illustration, we set the trading capital at 1 million dollars, opening and closing positions daily. Regarding transaction costs, we take account of the two main components. The first is the cost attached to the bid-ask spread. For the most liquid contract, the bid-ask spread can be as low as \$0.01 per barrel. In our case, we set the bid-ask spread at \$0.02 per barrel for generality. Since each Brent crude oil futures contract represents 1000 barrels, the bid-ask spread cost is $\$0.02 \times 1000 = \20 per contract in our case. An additional cost to consider is the exchange fee, which is set at \$1.5 per contract per trade. According to the ICE, the current applied margin rate for Brent crude oil varies between

1668.9 and 5310. Due to the highly leveraged nature of futures trading, we conservatively set the initial margin at \$5000 per contract.

Since the inception of the CAPM about 60 years ago, the performance of trading strategies has been assessed by their CAPM alpha. While it is generally acknowledged that the CAPM has clear limitations, even today, hundreds of hedge funds still assess and benchmark their strategies in terms of alpha, and salaries of fund managers are often determined through it. It is well known that in a basic setup like the Black-Scholes model, a strategy with a positive α is equivalent to the existence of an arbitrage. In more general models, specifically in the case of incompleteness, this no longer holds. Ewald and Taub (2022) link the CAPM to the so-called minimal martingale measure. However, alpha remains the most important performance measure, accounting for both a tradeoff between risk and return, and the possibility of diversifying (at least most of) the idiosyncratic risk. In this paper, we consider alpha as a performance measure that provides some evidence for a market anomaly, but not as a clear sign for an arbitrage. Nevertheless, this paper is not only about alpha and trading performance, it is as much about identifying an intra-day seasonal pattern which has not been known before. The way how we use alpha in our analysis establishes this seasonality independently and contributes to results contained through statistical testing in Section 2.



Figure 5. Intraday average price per hour across trading days.

As discussed in Ewald *et al.* (2022b), the leveraged nature of futures trading requires only a relatively small amount of capital to initiate trades, which makes it challenging to apply the traditional CAPM in the futures market. Therefore, an adjusted form of the CAPM that accounts for absolute instead of relative price changes is required. The appropriateness of this version of the CAPM is well established through classical literature, compare (e.g. Dusak 1973, Baxter *et al.* 1985):[†]

$$r_{t,i} = \alpha_i^* + \beta_i^* (r_{t,m} - r_t^f) + \epsilon_{t,i} \quad (1)$$

[†] One issue with the application of the CAPM in the context of financial futures is that because of an exact equal amount of long and short position (by definition), futures are in net-zero supply. However, the literature concludes that commodities are, in fact, implicitly included in the CAPM market portfolio through the firms that either produce or use the commodities in their manufacturing processes. Baxter *et al.* (1985) argues that explicitly accounting for commodities in a relevant market portfolio, for example, through including a relevant commodity index, would lead to double accounting of commodities. Ewald *et al.* (2022a) provide a more extensive overview of this. We therefore argue that the S&P500 is the appropriate index to benchmark our trading strategy against. This is also in alignment with the hundreds of hedge funds that assess their trading strategies, which in many cases involve commodity futures as well as other derivatives in net-zero supply, against the S&P500 benchmark or another broad equity-based index.

where $r_{t,i}$ represents the change in the futures price during a discrete time interval starting at time t , $r_{t,m}$ is the return of the market portfolio over the same period, r_t^f is the risk-free rate, α_i^* captures the excess return generated by trading a specific futures contract, β_i^* measures the sensitivity of the futures price change to the market portfolio, and $\epsilon_{t,i}$ is the error term.[‡]

For the single-contract strategy, we go long F1 at its lowest point, which occurs at 5:00 p.m., and go short at its highest point, at 11:00 a.m. The spread strategy, which involves trading both F1 and F3, is slightly more complex. The spread is defined as $F1 - F3$, and we still long the spread when the price is low (at 6:00 p.m.) and short the spread when the price is high (at 10:00 p.m.). Specifically, when we go long on the spread, we simultaneously go long F1 and go short F3; when we short the spread, we short F1 and long F3 at the same time. We repeat this trading pattern every day, and table 4 provides a brief overview of the overall performance as evaluated through the adjusted CAPM. The risk-free rate is the 3-month Treasury bill rate, and we use the S&P 500 as a proxy for the overall market performance. For both strategies, significant

[‡] Note that the scale of the regression coefficients α_i^* and β_i^* is affected by this transformation and they typically take larger values than in the classical version of the CAPM. However, while this is relevant for the interpretation of these coefficients, it isn't for the matter of statistical significance. In addition, a positive and significant α_i^* still represents abnormal returns and a violation to the CAPM.

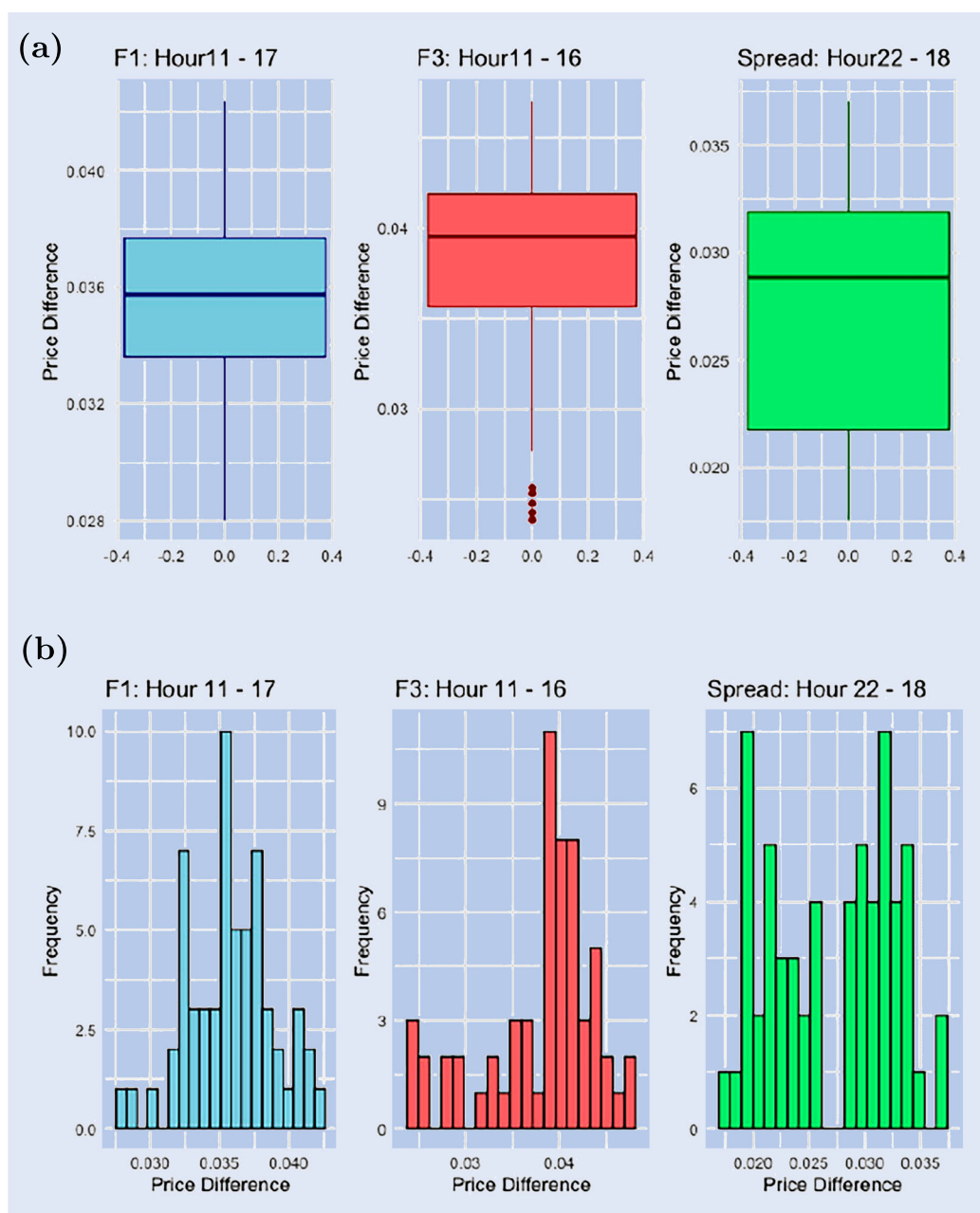


Figure 6. Visual inspection of intraday price differences to detect seasonal patterns: (a) Boxplot; (b) Histogram.

positive abnormal returns (α) are generated. The α for the F1 strategy is slightly higher than that obtained from the spread strategy. The β for the F1 strategy is negative, confirming the negative relationship between commodities, particularly Brent crude oil, and the stock market as documented in the classical literature. In contrast, the β for the spread strategy is positive, indicating that the combination of futures contracts may follow the stock market trend. As a result, we can confirm the profitability of the proposed trading strategy based on intra-day seasonality. The main rationale for the spread strategy is risk reduction, and in fact, all trades involved in this strategy are hedged at all times during the day, F1 against F3. This is very noticeable in the very low beta of that strategy; it carries extremely low systematic risk but still performs to a positive alpha.

Table 4. Trading performance evaluated by CAPM.

	Futures Return	
	F1 Strategy	Spread Strategy
α	0.04*** (0.02)	0.03*** (0.01)
β	-18.92*** (1.42)	2.56*** (0.65)
Observations	3018	3018
R^2	0.06	0.01

Notes: Standard errors in parentheses. [*] $p < 0.1$; [**] $p < 0.05$; [***] $p < 0.01$.

4. Trading performance

4.1. Single contract trading

The trading performance of the strategy based on F1 is depicted in figure 7, which illustrates the accumulated wealth over time both with and without transaction costs. In each case, we start with an initial wealth of \$1 million and consider three different scenarios. The first scenario assumes that the profit or loss from futures trading is simply set aside with no reinvestment—essentially holding the cash until the end. In the second scenario, the profit or loss is placed into 3-month Treasury bills, earning a risk-free rate. The third scenario involves reinvesting the profit or loss into the stock market, specifically

the S&P 500 index. As shown in the figure, the green line, which represents the cumulative wealth with reinvestment in the S&P 500, generates a higher level of wealth than the other two scenarios. Following this is the case with risk-free reinvestment, while the cash-holding scenario results in the lowest wealth accumulation. When accounting for transaction costs, total wealth is reduced, but still very considerable.

Table 5 summarizes the performance metrics of the F1 strategy. The total return represents the cumulative return generated through futures trading, while the total wealth is the sum of this return and the initial trading capital. The winning ratio is calculated as the proportion of profitable trades (i.e. trades with positive returns) relative to the total number of trades, providing a clear measure of the strategy’s success

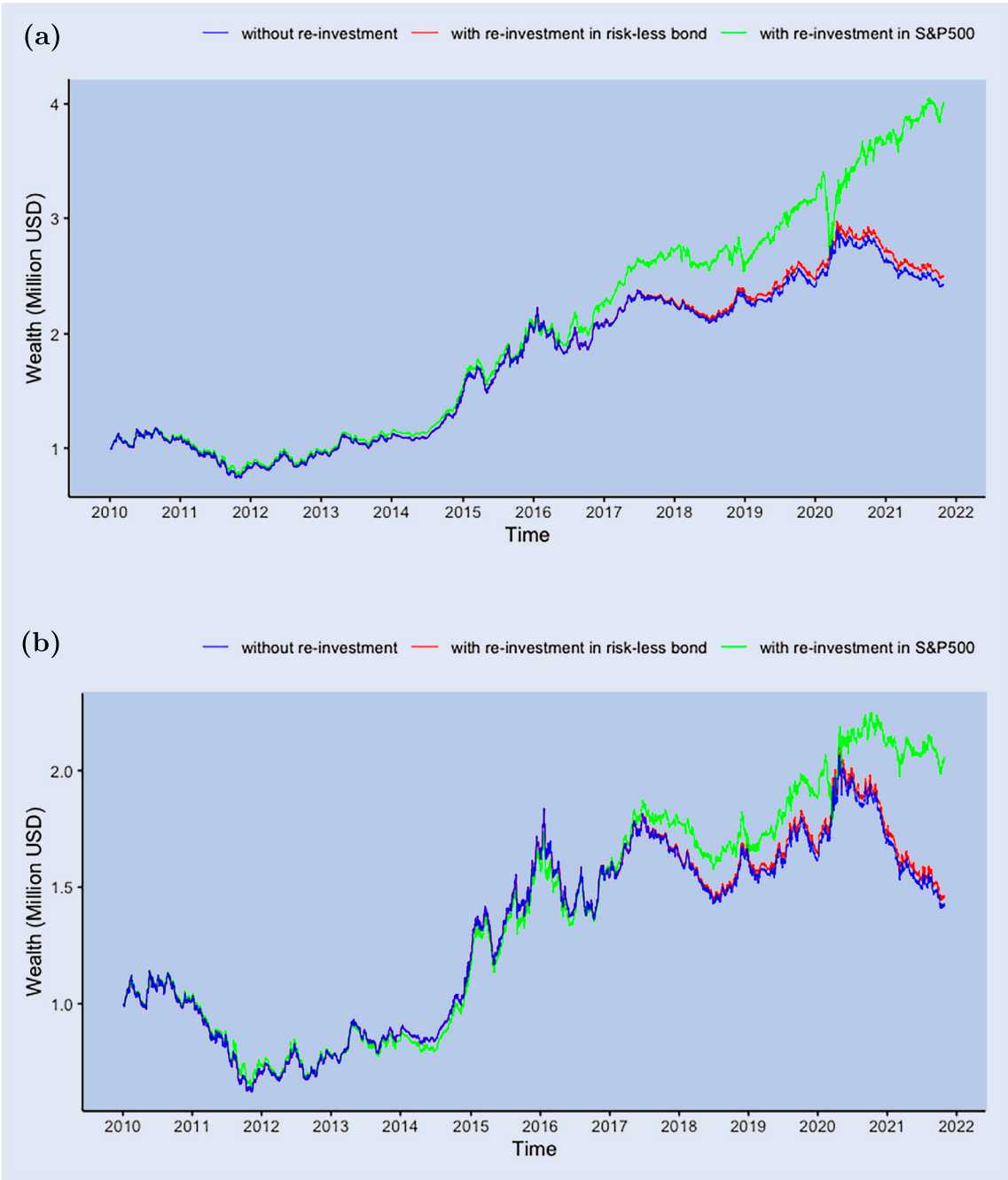


Figure 7. Cumulative wealth based on F1 strategy: (a) Without cost; (b) With cost.

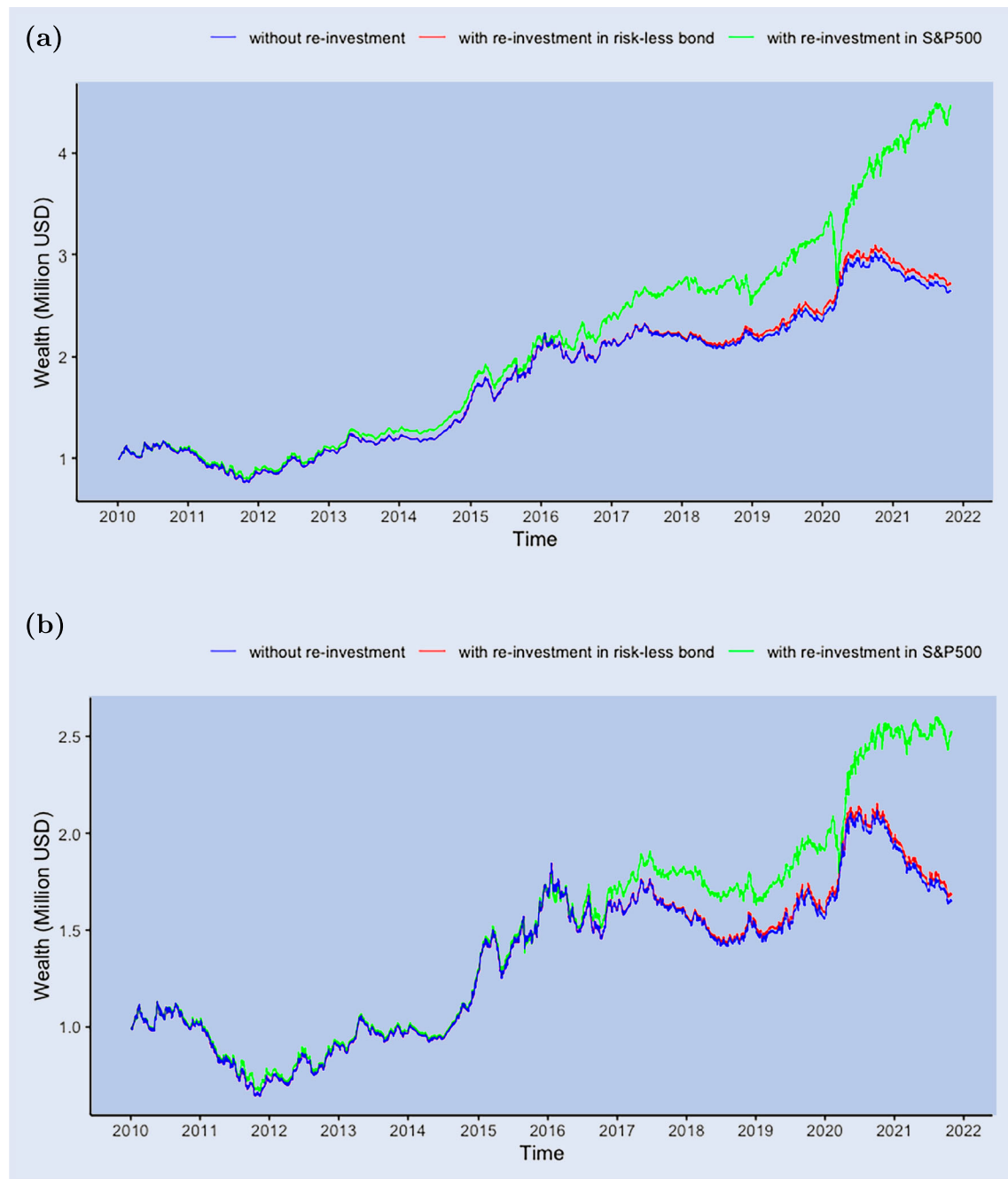


Figure 8. Cumulative wealth based on F3 strategy: (a) Without cost; (b) With cost.

rate by indicating the percentage of trades that result in a profit. The profit factor, defined as the ratio of gross profits to gross losses, evaluates the risk-reward balance of the trading strategy, with higher values indicating greater profitability. The annualized return on initial margin reflects performance adjusted for leverage, offering insights into the returns generated based on the margin used, and giving a more accurate picture of risk-adjusted returns. Additionally, the annualized return on trading capital reflects the raw performance relative to the initial trading capital, which is set at \$1 million in this case. The Sharpe ratio is calculated based on two different annualized rates of return to reflect the risk-adjusted performance. Overall, the F1 trading strategy demonstrated strong performance, even when accounting for transaction costs. Reinvesting into the S&P 500 can generate higher

returns, as it anchors our long-short arbitrage strategy against the benchmark index. Additionally, it is important to note that transaction costs play a significant role in this strategy due to the high trading frequency, implemented on a daily basis.

We also provide the trading performance based on the F3 contract, as shown in figure 8 and table 6. The F3 trading strategy outperforms the F1 strategy, both with and without transaction costs. In addition to the proposed trading strategy, two naïve benchmarks are considered for performance comparison: a pure investment in the risk-free bond and a buy-and-hold strategy in the S&P 500 index. The risk-free benchmark represents continuous investment in a low-volatility asset, yielding a total cumulative return of \$64,825, resulting in total wealth of \$1,064,825 and an annualized return

Table 5. Trading performance of F1 strategy.

	No Interest	3-Month Treasury Bill	S&P 500
<i>Without Cost</i>			
Total Return	1 432 215.7232	1 508 918.0109	3 019 832.1824
Total Wealth	2 432 215.7232	2 508 918.0109	4 019 832.1824
Winning Ratio	51.47%	51.51%	52.37%
Profit Factor	1.1093	1.1155	1.1675
Annualized Return (Initial Margin)	34.61%	34.96%	40.45%
Annualized Return (Trading Capital)	7.81%	8.09%	12.49%
Sharpe Ratio (Initial Margin)	0.6614	0.6922	0.8565
Sharpe Ratio (Trading Capital)	0.5466	0.5772	0.7433
<i>With Cost</i>			
Total Return	426 632.7232	462 462.0723	1 056 735.7172
Total Wealth	1 426 632.7232	1 462 462.0723	2 056 735.7172
Winning Ratio	50.08%	50.12%	50.31%
Profit Factor	1.0314	1.0341	1.0713
Annualized Return (Initial Margin)	28.67%	28.94%	32.71%
Annualized Return (Trading Capital)	3.05%	3.27%	6.29%
Sharpe Ratio (Initial Margin)	0.2356	0.2497	0.3829
Sharpe Ratio (Trading Capital)	0.1456	0.1599	0.3468

Note: The total return represents the cumulative return generated through futures trading, while the total wealth is the sum of this return and the initial trading capital. The winning ratio is calculated as the proportion of profitable trades (i.e. trades with positive returns) relative to the total number of trades, providing a clear measure of the strategy's success rate by indicating the percentage of trades that result in a profit. The profit factor, defined as the ratio of gross profits to gross losses, evaluates the risk-reward balance of the trading strategy, with higher values indicating greater profitability. The annualized return on initial margin reflects performance adjusted for leverage, providing insights into the returns generated based on the margin used and offering a clearer picture of risk-adjusted returns. Additionally, the annualized return on trading capital reflects the raw performance relative to the initial trading capital, which in this case is set at \$1 million. If the final wealth is negative, indicating a total loss in the strategy, the annualized return cannot be computed, and a dash (—) is used to represent the result. The Sharpe ratio is calculated based on two different annualized rates of return to reflect the risk-adjusted performance.

Table 6. Trading performance of F3 strategy.

	No Interest	3-Month Treasury Bill	S&P 500
<i>Without Cost</i>			
Total Return	1 652 068.8229	1 726 036.0096	3 479 535.6809
Total Wealth	2 652 068.8229	2 726 036.0096	4 479 535.6809
Winning Ratio	52.53%	52.63%	53.26%
Profit Factor	1.1415	1.1483	1.1994
Annualized Return (Initial Margin)	35.66%	35.98%	41.81%
Annualized Return (Trading Capital)	8.60%	8.85%	13.52%
Sharpe Ratio (Initial Margin)	0.7743	0.8070	0.9676
Sharpe Ratio (Trading Capital)	0.7291	0.7630	0.8738
<i>With Cost</i>			
Total Return	652 442.8229	685 722.4179	1 527 191.6779
Total Wealth	1 652 442.8229	1 685 722.4179	2 527 191.6779
Winning Ratio	51.01%	51.11%	51.51%
Profit Factor	1.0537	1.0565	1.1100
Annualized Return (Initial Margin)	30.34%	30.56%	35.11%
Annualized Return (Trading Capital)	4.34%	4.52%	8.16%
Sharpe Ratio (Initial Margin)	0.3049	0.3193	0.5438
Sharpe Ratio (Trading Capital)	0.2709	0.2862	0.5474

Note: The total return represents the cumulative return generated through futures trading, while the total wealth is the sum of this return and the initial trading capital. The winning ratio is calculated as the proportion of profitable trades (i.e. trades with positive returns) relative to the total number of trades, providing a clear measure of the strategy's success rate by indicating the percentage of trades that result in a profit. The profit factor, defined as the ratio of gross profits to gross losses, evaluates the risk-reward balance of the trading strategy, with higher values indicating greater profitability. The annualized return on initial margin reflects performance adjusted for leverage, providing insights into the returns generated based on the margin used and offering a clearer picture of risk-adjusted returns. Additionally, the annualized return on trading capital reflects the raw performance relative to the initial trading capital, which in this case is set at \$1 million. If the final wealth is negative, indicating a total loss in the strategy, the annualized return cannot be computed, and a dash (—) is used to represent the result. The Sharpe ratio is calculated based on two different annualized rates of return to reflect the risk-adjusted performance.

of 0.53%. While the return is modest, it exhibits remarkable stability, with nearly all trading days generating a positive return. By contrast, the S&P 500 benchmark reflects a long-term passive exposure to the equity market. It achieves a significantly higher cumulative return of \$3,064,802 and total wealth of \$4,064,802, corresponding to an annualized return of 12.59%. However, this comes with greater volatility, as evidenced by a Sharpe ratio of 0.698, a winning ratio of 55.21% and a profit factor of 1.165. These two benchmarks serve as useful references to contextualize the performance and risk profile of the intraday trading strategy. In general, both the F1 and F3 strategies outperform the benchmarks and demonstrate relatively consistent performance. However, once transaction costs are taken into account, their performance in terms of the Sharpe ratio lags behind that of the S&P 500 index. However, as is well known, the Sharpe ratio does not distinguish between systematic and idiosyncratic risk in the futures trading strategy, which can be diversified (at least to some extent) by other means. The latter is accounted for by the CAPM alpha, which ways in favor of our proposed strategies.

4.2. Spread trading

The spread trading strategy is similar to the single contract strategy, except that it involves two futures contracts at each point in time, specifically F1 and F3 in our case. When the spread is high, we short the spread, meaning we short F1 and simultaneously long F3. Conversely, when the spread is low, we take the opposite position by going long on F1 and short on F3. We evaluate the performance under three different scenarios (cash holding, risk-free investment, and stock index investment), assuming an initial trading capital of \$1 million every day. The trading performance of the spread strategy, both with and without transaction costs, is shown in figure 9. As depicted in the figure, without transaction costs, the green line—representing cumulative wealth with reinvestment in the S&P 500—generates higher wealth compared to the other two scenarios. This is followed by risk-free reinvestment, while the cash-holding scenario results in the lowest wealth accumulation. However, when transaction costs are considered, the situation reverses. The strategy incurs total losses in all cases, with the cash-holding scenario suffering the least, while reinvesting in the stock market exacerbates

Table 7. Trading performance of spread strategy.

	No Interest	3-Month Treasury Bill	S&P 500
<i>Without Cost</i>			
Total Return	727 958.5432	769 759.9098	1 874 541.7817
Total Wealth	1 727 958.5432	1 769 759.9098	2 874 541.7817
Winning Ratio	52.89%	53.66%	55.98%
Profit Factor	1.2072	1.2203	1.2123
Annualized Return (Initial Margin)	23.32%	23.57%	28.74%
Annualized Return (Trading Capital)	4.73%	4.95%	9.34%
Sharpe Ratio (Initial Margin)	1.1173	1.1555	1.2343
Sharpe Ratio (Trading Capital)	0.6597	0.7012	0.8840
<i>With Cost</i>			
Total Return	− 1 284 633.4568	− 1 324 624.6573	− 2 054 190.1097
Total Wealth	− 284 633.4568	− 324 624.6573	− 1 054 190.1097
Winning Ratio	34.61%	34.42%	42.40%
Profit Factor	0.7277	0.7211	0.7793
Annualized Return (Initial Margin)	–	–	–
Annualized Return (Trading Capital)	–	–	–
Sharpe Ratio (Initial Margin)	− 0.7871	− 0.8238	− 0.8547
Sharpe Ratio (Trading Capital)	− 1.3297	− 1.3689	− 1.0499

Note: The total return represents the cumulative return generated through futures trading, while the total wealth is the sum of this return and the initial trading capital. The winning ratio is calculated as the proportion of profitable trades (i.e. trades with positive returns) relative to the total number of trades, providing a clear measure of the strategy's success rate by indicating the percentage of trades that result in a profit. The profit factor, defined as the ratio of gross profits to gross losses, evaluates the risk-reward balance of the trading strategy, with higher values indicating greater profitability. The annualized return on initial margin reflects performance adjusted for leverage, providing insights into the returns generated based on the margin used and offering a clearer picture of risk-adjusted returns. Additionally, the annualized return on trading capital reflects the raw performance relative to the initial trading capital, which in this case is set at \$1 million. If the final wealth is negative, indicating a total loss in the strategy, the annualized return cannot be computed, and a dash (–) is used to represent the result. The Sharpe ratio is calculated based on two different annualized rates of return to reflect the risk-adjusted performance.

the losses and performs the worst. The transaction costs associated with spread trading are particularly high, as each trade involves two contracts, and both positions must be opened and closed within a day. Therefore, even when profitable opportunities exist without considering transaction costs, as shown in plot (a) of figure 9, the profits can be significantly eroded by the costs inherent in high-frequency trading. In conclusion, the spread strategy can only be recommended when substantial discounts for trading costs are offered, for example, for institutional investors. On the other hand, the spread strategy has much lower risk than the single contract strategies due to simultaneous long-short trading, and hence we see an example where transaction costs incentivise risk-taking.

Table 7 summarizes the performance metrics of the spread strategy. In the absence of transaction costs, the strategy performs well. Although its total return, total wealth, and annualized return are lower compared to the single contract strategy, its winning ratio, profit factor, and Sharpe ratio are all higher. Notably, the Sharpe ratio is greater than 1.2, indicating that spread trading generates a much more stable risk-adjusted return than the single contract strategy. However, once transaction costs are accounted for, the strategy incurs significant losses. Negative returns and wealth are observed,

Table 8. Trading performance of spread strategy: adjusted cost as single contract.

	No Interest	3-Month Treasury Bill	S&P 500
Total Return	− 278 337.4568	− 277 432.3738	− 89 824.1640
Total Wealth	721 662.5432	722 567.6262	910 175.8360
Winning Ratio	41.44%	41.44%	47.07%
Profit Factor	0.9315	0.9317	0.9804
Annualized Return (Initial Margin)	14.54%	14.55%	16.81%
Annualized Return (Trading Capital)	− 2.72%	− 2.71%	− 0.79%
Sharpe Ratio (Initial Margin)	0.1651	0.1657	0.3541
Sharpe Ratio (Trading Capital)	− 0.3363	− 0.3354	− 0.1416

Note: The total return represents the cumulative return generated through futures trading, while the total wealth is the sum of this return and the initial trading capital. The winning ratio is calculated as the proportion of profitable trades (i.e. trades with positive returns) relative to the total number of trades, providing a clear measure of the strategy's success rate by indicating the percentage of trades that result in a profit. The profit factor, defined as the ratio of gross profits to gross losses, evaluates the risk-reward balance of the trading strategy, with higher values indicating greater profitability. The annualized return on initial margin reflects performance adjusted for leverage, providing insights into the returns generated based on the margin used and offering a clearer picture of risk-adjusted returns. Additionally, the annualized return on trading capital reflects the raw performance relative to the initial trading capital, which in this case is set at \$1 million. If the final wealth is negative, indicating a total loss in the strategy, the annualized return cannot be computed, and a dash (–) is used to represent the result. The Sharpe ratio is calculated based on two different annualized rates of return to reflect the risk-adjusted performance.

with very low winning ratios and profit factors. Additionally, the Sharpe ratio becomes negative, all of which indicates poor profitability. It is important to note that when the final wealth is negative, signifying a total loss in the strategy, the annualized return cannot be computed, and a dash (–) is used to represent this result. Overall, the spread trading strategy demonstrates strong performance when transaction costs are excluded, with reinvestment into the S&P 500 yielding higher returns. However, when costs are factored in, profits are erased, leading to significant losses. Once again, it is crucial to acknowledge that transaction costs play a significant role, especially given the high-frequency nature of intra-day trading.

To better understand the effect of trading costs, we further evaluate the performance of the spread trading strategy by calculating the trading costs as if they were incurred for a single contract rather than doubling the costs to account for two separate contracts in the spread. This is, in fact, a reasonable assumption as the ICE offers various discounts on block trades. The cumulative wealth under this adjusted cost structure is presented in figure 10, with additional performance measures summarized in table 8. It is observed that the trading strategy demonstrates relatively strong performance up until 2015, after which a sharp decline becomes evident. This decline coincides with a significant drop in oil prices, as shown in figure 3, highlighting the potential influence of market conditions on trading performance and the importance of cost assumptions in evaluating strategy outcomes. In a more advanced strategy, longer-term systematic price risk could be hedged by including longer-term maturities the strategy. However, we leave this for future research.

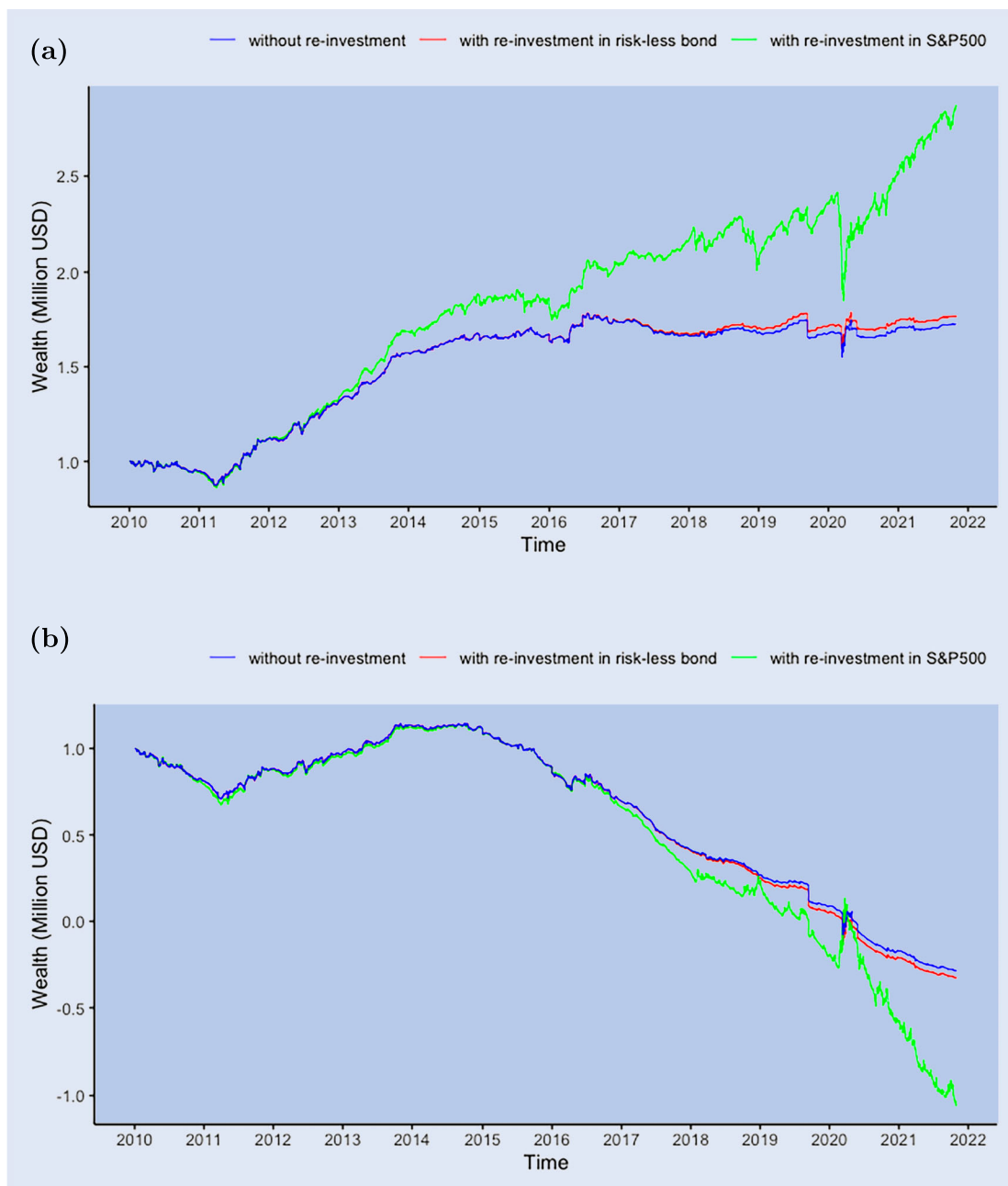


Figure 9. Cumulative wealth based on spread strategy: (a) Without cost; (b) With cost.

5. Conclusions

We have shown that intra-day seasonal patterns in Brent Crude oil futures prices traded at the ICE exist, they are statistically significant, and to an extent that they can be exploited through appropriately designed arbitrage strategies. More specifically, using a raw data set of 22 terabytes (TB) of Commodities Level 1 data from the Intercontinental Exchange in London, spanning the period from January 2010 until October 2021, containing tick-by-tick data of Brent crude oil futures contracts, we showed that futures prices are statistically significantly higher at certain times of the day than at other times of the day and that this intra-day seasonal pattern can be facilitated to design appropriate long-short and spread strategies

that can produce abnormal returns in terms of positive and significant CAPM alphas, even when accounting for reasonable transaction costs. To the best of our knowledge, our study is the first to establish intra-day seasonal patterns in a key commodity, crude oil, and the first to look at the potential of intra-day seasonal patterns in asset prices for the creation of abnormal returns and arbitrage. As such, our paper makes significant contributions to the topics of seasonality and arbitrage in commodity prices and, more generally, to the topics of commodity futures trading and financial market microstructure. In future research, we plan to tie our study to liquidity data, aiming to understand the impact of liquidity risk premia over an intra-day cycle, which may shed light on the underlying reason for the intra-day seasonal pattern.

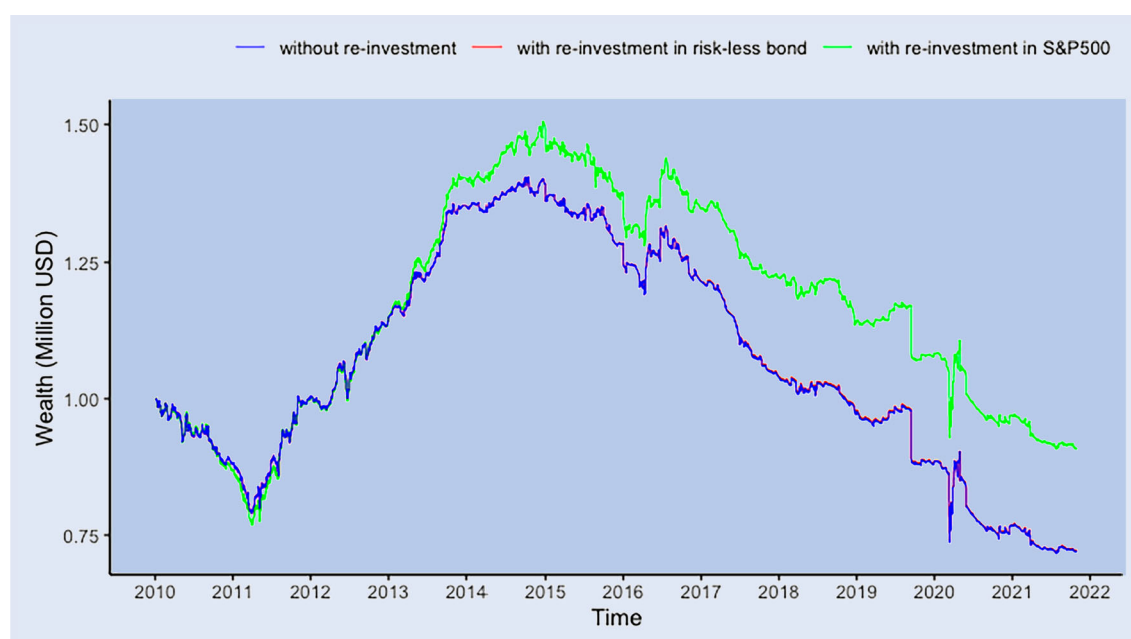


Figure 10. Cumulative wealth based on spread strategy: adjusted cost as single contract.

Note: The trading costs in this figure are calculated under the assumption that costs are incurred for a single contract, rather than being doubled to reflect two separate contracts in a typical spread trade. This is a reasonable assumption, as the Intercontinental Exchange (ICE) offers various discounts on block trades, effectively reducing the cost burden for spread positions.

Disclosure statement

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References

- Ahmed, W.S., Sohaib, M., Maqsood, J. and Siddiqui, A., Do intraday week effect in currencies hourly trading reflect leverage and asymmetric anomalies? Policy implications for traders. *J. Chin. Econ. Foreign Trade Stud.*, 2021, **14**(1), 240–256.
- Aradhyula, S.V. and Ergün, A.T., Trading collar, intraday periodicity and stock market volatility. *Appl. Financ. Econ.*, 2004, **14**(13), 909–913.
- Baxter, J., Conine, T.E. and Tamarkin, M., On commodity market risk premiums: Additional evidence. *J. Futures Mark.*, 1985, **5**(1), 121–125.
- Blau, B.M., Van Ness, B.F. and Van Ness, R.A., Intraday stealth trading: Which trades move prices during periods of high volume? *J. Financ. Res.*, 2009, **32**(1), 1–21.
- Bohl, M.T., Schuppli, M. and Siklos, P.L., Stock return seasonalities and investor structure: Evidence from China's B-Share markets. *China Econ. Rev.*, 2010, **21**(1), 190–201.
- Diao, X. and Tong, B., Forecasting intraday volatility and VaR using multiplicative component GARCH model. *Appl. Econ. Lett.*, 2015, **22**(18), 1457–1464.
- Dusak, K., Futures trading and investor returns: An investigation of commodity market risk premiums. *J. Political Econ.*, 1973, **81**(6), 1387–1406.
- Enow, S.T., Revisiting the January effect anomaly: Evidence from international stock markets. *Int. J. Res. Bus. Soc. Sci.* (2147–4478), 2024, **13**(4), 245–251.
- Ewald, C.O., Haugom, E., Kanthan, L., Lien, G., Salehi, P. and Størdal, S., Salmon futures and the fish pool market in the context of the CAPM and a three-factor model. *Aquac. Econ. Manag.*, 2022a, **26**(2), 171–191.
- Ewald, C.O., Haugom, E., Lien, G., Størdal, S. and Wu, Y., Trading time seasonality in commodity futures: An opportunity for arbitrage in the natural gas and crude oil markets? *Energy Econ.*, 2022b, **115**, 106324.
- Ewald, C.O. and Taub, B., Real options, risk aversion and markets: A corporate finance perspective. *J. Corp. Finance*, 2022, **72**, 102164.
- Faff, R.W. and McKenzie, M.D., The impact of stock index futures trading on daily returns seasonality: A multicountry study. *J. Bus.*, 2002, **75**(1), 95–126.
- Giot, P., Time transformations, intraday data, and volatility models. *J. Comput. Finance*, 2000, **4**(2), 31–62.
- Giot, P., Market risk models for intraday data. *Eur. J. Finance*, 2005, **11**(4), 309–324.
- Hansen, P.R. and Lunde, A., Realized variance and market microstructure noise. *J. Bus. Econ. Stat.*, 2006, **24**(2), 127–161.
- Hsieh, T.Y., Fu, Y.F. and Ya, S., Impacts of day trading on the intraday pattern of market quality. *Int. J. Serv. Technol. Manag.*, 2020, **26**(1), 20–37.
- Oprea, D.S. and Țilică, E.V., Day-of-the-week effect in post-communist east European stock markets. *Int. J. Acad. Res. Account. Finance Manag. Sci.*, 2014, **4**(3), 119–129.
- Ryu, D., Intraday price formation and bid-ask spread components: A new approach using a cross-market model. *J. Futures Mark.*, 2011, **31**(12), 1142–1169.
- Saad, M. and Darrat, A.F., Intraday pattern in liquidity covariation: Evidence from NYSE listed firms. *Appl. Financ. Econ.*, 2007, **17**(11), 913–919.
- Siklos, P.L., Stock return seasonalities and investor structure: Evidence from China's B-Share market. *SSRN Electron. J.*, 2008.

- Størdal, S., Ewald, C.O., Lien, G. and Haugom, E., Trading time seasonality in electricity futures. *J. Commod. Mark.*, 2023, **31**, 100291.
- Tannous, G.F., Wang, J. and Wilson, C., The intraday pattern of information asymmetry, spread, and depth: Evidence from the NYSE. *Int. Rev. Finance*, 2013, **13**(2), 215–240.
- Tian, G.G. and Guo, M., Interday and intraday volatility: Additional evidence from the shanghai stock exchange. *Rev. Quant. Finance Account.*, 2007, **28**(3), 287–306.
- Tissaoui, K., The intraday pattern of trading activity, return volatility and liquidity: Evidence from the emerging Tunisian stock exchange. *Int. J. Econ. Finance*, 2012, **4**(6), 171–181.
- Visaltanachoti, N., Charoenwong, C. and Ding, D.K., Spread, depth, and order flow patterns of warrants and their underlying stocks on the stock exchange of Thailand. *SSRN Electron. J.*, 2003.
- Xu, Y., Bouri, E., Saeed, T. and Wen, Z., Intraday return predictability: Evidence from commodity ETFs and their related volatility indices. *Resour. Policy*, 2020, **68**, 101800.
- Xu, Y. and Li, C., Liquidity of the Chinese agricultural futures market and its impact on futures price—based on high-frequency data. *Sustainability*, 2018, **10**(11), 3992.